

Deepika Patibandla

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Summary

Data Scientist with **over 4 years of experience** applying machine learning, NLP, and AI-driven analytics across healthcare, finance, and enterprise domains. Hands-on with **Python, SQL, and cloud-based ML pipelines (AWS, Azure)** to solve real business problems like automating claim reviews, predicting defaults, and detecting equipment failure. Skilled at building **retrieval-augmented generation (RAG) workflows**, fine-tuning LLMs (LLaMA-2, Mistral), and deploying end-to-end models that cut manual effort, improve accuracy, and support data-driven decisions.

Experience

United Health Group | AI/ML Engineer

Jan 2025 - Current

- Built a **RAG-based claim summarization system** using LangChain, FAISS, and Hugging Face embeddings to help analysts search and interpret 10M+ medical notes; reduced manual lookup time by 40%.
- Fine-tuned **LLaMA-2** and **Mistral** models on de-identified patient data to automate prior-auth case summaries, improving accuracy and compliance by 20%.
- Developed **feature pipelines in Python and SQL** to clean and merge 50M+ claim records; boosted model throughput and lowered preprocessing latency by 60%.
- Created **evaluation dashboards in Power BI** to track precision, recall, and drift metrics, helping product leads monitor model quality in production.
- Deployed end-to-end ML workflows on **AWS (Lambda, ECS, Glue)** with CI/CD automation, ensuring HIPAA-aligned, reproducible model environments.

Deloitte | Associate Software Engineer - Data Science Team

Jan 2022 - Jul 2023

- Designed **fraud-detection models** in Python (XGBoost, Isolation Forest) analyzing 100M+ daily transactions; raised fraud catch rate by 24% while cutting false alerts.
- Automated **ETL pipelines** in SQL and Pandas to transform financial data from multiple sources; shortened refresh cycles by 30%.
- Built **QlikSense and Power BI dashboards** to visualize fraud trends and KPI drift, giving compliance teams real-time monitoring across business units.
- Tuned and validated models using **A/B testing** and hyperparameter optimization, increasing model precision by 18%.
- Containerized and deployed inference services via **AWS ECS and Docker**, reducing deployment time by 40% and ensuring reproducibility.

7fincorp Group | Data Scientist

Feb 2020 - Dec 2021

- Developed **Python-based data pipelines** to process 20M+ customer activity logs for churn and anomaly detection; cut data prep time by 40%.
- Built ML models (Random Forest, KMeans, TensorFlow) for churn prediction and segmentation, increasing retention-model accuracy by 22%.
- Designed **NLP workflows** using Hugging Face and spaCy to classify 5M+ support tickets, reducing manual review by 30%.
- Integrated **SQL aggregation and ML pipelines** to unify structured and text data, accelerating experimentation by 35%.
- Collaborated with business teams to translate model insights into marketing and retention strategies that improved renewal rates quarter-over-quarter.

Technical Skills

Machine Learning & Data Science: Scikit-learn, XGBoost, Random Forest, Regression & Classification Models, Clustering (KMeans), Time Series Forecasting, Recommender Systems, Anomaly Detection, Computer Vision, Feature Engineering, EDA, Statistical Modeling, Hypothesis Testing, Experimentation, ANOVA

AI & Deep Learning / LLM: Hugging Face Transformers, LangChain, LLaMA-2, Mistral, GPT-4, RAG Pipelines, Prompt Engineering, FAISS, Pinecone, TensorFlow, PyTorch, Keras, NLP (embeddings, tokenization, transformer-based models)

Data Engineering & Cloud: MySQL, SQL Server, Oracle, PostgreSQL, MongoDB, Amazon Services, Microsoft Azure

Data Visualization & Analytics: Matplotlib, Seaborn, Tableau, Power BI, QlikSense, Quick Sight, Looker, Microsoft Excel, KPI Reporting, Predictive Modeling

Programming & Tools: Python (NumPy, Pandas, SciPy), SQL, Shell Scripting, Git, Docker, Kubernetes, CI/CD Pipelines, Statistical Analysis, Statistical Inference, Financial, Quantitative Analysis

Certifications: Certified Scrum Master | Python - NPTEL | OCI – AI Foundations

Education

North Carolina State University | Masters in Computer Science (**Data Science Track**) | GPA 4.0/4.0 | May 2025